Karyampudi Meghana Sai

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.01
Sri Chaitanya Junior College, Guntur	Class XII (Board of Intermediate Education Andhra Pradesh)	2023	97.40%
Sri Chaitanya High School, Guntur	Class X (Board of Secondary Education Andhra Pradesh)	2021	98.50%

EXPERIENCE

ORBCOMM Technologies India Pvt. Ltd. | AI Intern

- Extended Shipment APIs with **customer shipment identifiers**, enforcing input validation, tenant-scoped uniqueness, and immutability; added unit and integration tests to validate API behavior and identifier propagation across data pipelines.
- Enhanced Bronze-Silver-Gold Medallion pipelines for Where Is My Container (WIMC) using **PySpark and Microsoft Fabric**, propagating shipment identifiers and asset metadata to support downstream shipment analytics.
- Developed and evaluated city/state extraction across **10,475 facility-address records**, achieving **76.0% city and 77.7% state accuracy** against reference fields to support structured shipment-location data.

PROJECTS

Vision Foundation Models for Dermatology | *Independent Project under **Prof. Sumohana Channappayya***

- Benchmarked CNN, ResNet18**,** and Swin Transformers on **HAM10000** and **ISIC 2019**, extending ISIC experiments with fine-tuned DINOv2 and MambaVision-T.
- Applied **progressive unfreezing,** class-weighted/focal losses**, MixUp**, and **EMA**; built a **DINOv2-MambaVision-T ensemble** with weighted soft voting and **4-view TTA**, recording **87.79%** accuracy and **0.8565** macro-F1.
- Analyzed errors using **Grad-CAM, t-SNE,** and confusion matrices, identifying persistent **melanoma-nevus confusion**; assessed accuracy, macro-F1, and inference latency.

TrustRAG

- Built TrustRAG, a **self-correcting Agentic RAG** system using LangGraph, Qdrant Cloud, Gemini embeddings, and LangChain, integrating BM25 + dense retrieval, Reciprocal Rank Fusion (RRF), and LLM reranking for evidence retrieval.
- Implemented **claim-level evidence verification**, bounded self-correction, re-verification, and **abstention** to address unsupported answers, with evidence-support scoring, source citations, and an interactive Streamlit interface.
- Evaluated on Kubernetes documentation against Basic RAG using a 30-question benchmark, achieving **100% vs. 20% abstention accuracy** on unanswerable queries, while measuring source-hit rates, correction frequency, and latency trade-offs.

BeaconFlow - AI-Powered Notification Orchestration Platform

- Engineered a notification platform using Next.js, NestJS, Prisma, and PostgreSQL, enabling scheduled and event-driven emails, reusable templates, and **persistent delivery tracking**.
- Architected a **producer-consumer pipeline** using BullMQ, Upstash Redis, and background workers, decoupling API requests from email delivery to support asynchronous processing, delayed jobs, and **automatic retries**.
- Integrated Google Gemini for **AI-assisted** email generation and refinement and Resend for delivery; deployed the full-stack application across Vercel, Render, Neon PostgreSQL, and Upstash Redis.

Compressed Sensing for Stock Market Data Recovery

- Implemented **Robust Matrix Completion** using nuclear-norm and L1 regularization using SCS solver to recover missing returns for 50 S&P 500 stocks, reducing **RMSE** by **6.0%** and **MAE** by **8.9%** versus KNN at 15% missing

data; improved portfolio Sharpe ratio from -0.640 to -0.538 versus KNN in an out-of-sample 2008 backtest.

Norm Classifier

- Built a cultural-norm classifier using CultureBank, Gemini-generated negatives, entity masking, and **adversarial filtering**; fine-tuned ELECTRA, DeBERTa-v3, and DistilBERT, achieving 98% in-domain accuracy with ELECTRA and **82% on 100 adversarial examples** with DistilBERT, versus 62% for BiLSTM.

Early Sepsis Detection Using Machine Learning

- Developed ML classifiers for sepsis detection on PhysioNet ICU data, achieving 0.83 ROC-AUC with XGBoost; implemented an **LSTM for six-hour-ahead prediction**, achieving **0.852 ROC-AUC** by modeling temporal patterns in patient vitals and laboratory measurements.

POSITIONS OF RESPONSIBILITY

IR Cell, IITH | *Web Coordinator* - Managed the IR Cell website, updating MOUs, events, exchange records, and international announcements in coordination with the IR Office.

Lambda, IITH | *Web Coordinator* - Helped with project ideas and supported ongoing tech projects for the IITH community.

RELEVANT COURSES

CS & ML Courses: Data Structures & Algorithms, Introduction to Programming, Introduction to AI/ML, Machine Learning, NLP, Predictive Modelling.

Math & Statistics Courses: Calculus, Differential Equations, Matrix Theory, Probability and Random Processes, Statistical Inference, Convex Optimization.

SKILLS

Languages & AI/ML: Python, C++, C, SQL; PyTorch, TensorFlow/Keras, scikit-learn, LangChain, LangGraph.

Backend & Tools: Next.js, NestJS, PySpark, PostgreSQL, Redis, Qdrant, GraphQL, Git, Docker, Microsoft Fabric.

EXTRACURRICULAR

Member, **National Sports Organization** Volleyball